

Homework Set 5

Propagator as Green's function:

- 1) Show that both the Advanced & retarded propagators are Green functions to the Klein Gordon operator
- 2) Show that the Feynman Propagator is a Green function for the Klein-Gordon operator.

Causality & Propagator

- 1) Calculate the two point function of massive free fields in $d+1$ dimensions
(hint: you might need Bessel functions)
- 2) Calculate the commutator.

Charge conjugation

Derive the action of the charge conjugation operator on a Dirac spinor.

Gordon identities

Derive the identities

$$\bar{u}_{s'}(\vec{p}') \left((p' + p)^\mu - 2i S^{\mu\nu} (p' - p)_\nu \right) \gamma_5 u_s(\vec{p}) = 0$$

$$\bar{v}_{s'}(\vec{p}') \left((p' + p)^\mu - 2i S^{\mu\nu} (p' - p)_\nu \right) \gamma_5 v_s(\vec{p}) = 0$$